Finite-dimensional Banach spaces satisfy Property (P)

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2026-06-19

Source question and scope

This packet concerns arXiv:2407.07490, Debmalya Sain, Arpita Mal, Kalidas Mandal and Kallol Paul, *On uniform Bishop-Phelps-Bollobas type approximations of linear operators and preservation of geometric properties*. The source paper works over the real scalar field and asks in Section 3:

> Is it true that each finite-dimensional Banach space X satisfies Property (P)? What about the infinite-dimensional case?

either $\|x_0 - h_0\| < \epsilon$ or $\|x_0 + h_0\| < \epsilon$, which contradicts the choice of ϵ. Therefore, the pair $(\mathbb{X}, \mathbb{X})$ does not have the uniform sBPBp with respect to $\mathcal{F}$. □

We end this section with an open question that we could not answer. In view of the results obtained here, it is apparent that an answer to the following question will be useful in having a better understanding of the uniform $\epsilon-$BPB approximations of operators as well as the norm attainment set of an operator.

Question: Is it true that each finite-dimensional Banach space $\mathbb{X}$ satisfies Property (P)? What about the infinite-dimensional case?

4. Applications of uniform $\epsilon-$BPB approximation on some particular Banach spaces

In this section we deal with operators on some well studied Banach spaces like ℓ_∞^n and ℓ_1^n, in view of the preservation of the norm attainment set under uniform $\epsilon-$BPB approximations. We note that in case of norm one operators which are not extreme contractions, the situation is completely understood by virtue of Theorem 3.4. Therefore, we restrict ourselves mostly to the case of extreme contractions. We begin with extreme contractions in $\mathbb{L}(\ell_\infty^n, \ell_\infty^n)$. First we require the following

The theorem below gives a full affirmative answer to the finite-dimensional part. It does not settle the infinite-dimensional part.

Definitions

Let X be a real Banach space. For $A \in \mathcal{L}(X)$, set

$$M_A = \{x \in S_X : \|Ax\| = \|A\|\}.$$

The space X has *Property (P)* if there is an $r_0 > 0$ such that for every norm-one non-isometry $A \in \mathcal{L}(X) \setminus \operatorname{Iso}(X)$, there is $x_A \in S_X$ with

$$B(x_A, r_0) \cap M_A = \varnothing.$$

Thus a failure of Property (P) in finite dimension is exactly the existence of norm-one non-isometries whose norm-attainment sets are arbitrarily dense in S_X.

For $x \neq 0$, write

$$J(x) = \{f \in S_{X^*} : f(x) = \|x\|\}.$$

These are the supporting functionals of the norm at x.

Proof intuition

Assume Property (P) fails. Then there are norm-one non-isometries A_n whose norm-attainment sets are $1/n$-dense in the unit sphere. Such operators almost preserve the norm of every unit vector, so finite-dimensional compactness forces them close to the isometry group. After conjugating by a nearest isometry, we get contractions

$$B_n = I + t_n C_n, \qquad t_n = \|B_n - I\| > 0, \qquad C_n \to C.$$

The core point is first-order but does not require smoothness. Fix $x \in S_X$. Contractivity of B_n implies $f(Cx) \leq 0$ for every supporting functional $f \in J(x)$. On the other hand, exact norm attainment at points $y_n \to x$ gives supporting functionals converging to some $g \in J(x)$ with $g(Cx) \geq 0$. Hence

$$\max_{f \in J(x)} f(Cx) = 0 \qquad (x \in S_X).$$

This says that the right derivative of $t \mapsto \left\|e^{tC} z\right\|$ is zero at every time and for every $z \neq 0$. Therefore e^{tC} is an isometry for all $t \geq 0$. But then the isometry $e^{t_n C}$ is closer to B_n than the chosen nearest isometry, a contradiction.

Two elementary lemmas

Lemma 1 (Directional derivative of the norm)**.** *Let X be a finite-dimensional real Banach space. If $x \neq 0$ and $v \in X$, then*

$$\lim_{t \downarrow 0} \frac{\|x + tv\| - \|x\|}{t} = \max_{f \in J(x)} f(v).$$

Proof. For every $f \in J(x)$,

$$\|x + tv\| - \|x\| \geq f(x + tv) - f(x) = tf(v),$$

so the lower limit is at least $\max_{f \in J(x)} f(v)$.

For the reverse inequality, choose $f_t \in J(x + tv)$. Then

$$\|x + tv\| - \|x\| = f_t(x + tv) - \|x\| \leq f_t(x + tv) - f_t(x) = tf_t(v).$$

Along any sequence $t_k \downarrow 0$, compactness of S_{X^*} gives a subsequence $f_{t_k} \to f$. Since $x + t_k v \to x$, the limit satisfies $f \in J(x)$. Taking upper limits gives the desired inequality. ☐

Lemma 2 (Zero right derivative)**.** *Let $h : [0, T] \to \mathbb{R}$ be continuous and suppose the right derivative*

$$h'_+(s) = \lim_{t \downarrow 0} \frac{h(s + t) - h(s)}{t}$$

exists and is zero for every $0 \leq s < T$. Then h is constant.

Proof. Fix $\varepsilon > 0$. For each $s < T$, the assumption $h'_+(s) = 0$ gives a $\delta_s > 0$ such that

$$|h(s+u) - h(s)| \le \varepsilon u$$

whenever $0 < u < \delta_s$ and $s + u \le T$. Let

$$E = \{r \in [0,T] : |h(t) - h(0)| \le \varepsilon t \text{ for every } 0 \le t \le r\}.$$

The set E is nonempty and closed by continuity of h. If $a = \sup E < T$, then $a \in E$, and the estimate at $s = a$ extends the same inequality to all $t \in [a, a + \delta_a)$, contradicting the definition of a. Hence $T \in E$, so $|h(T) - h(0)| \le \varepsilon T$. Since ε is arbitrary, $h(T) = h(0)$. Applying the same argument on every subinterval $[a,b] \subset [0,T]$ shows that h is constant. $\square$

Main theorem

Theorem 1. *Every finite-dimensional real Banach space satisfies Property* (P).

Proof. Let X be finite-dimensional, and denote by $G = \mathrm{Iso}(X)$ the group of surjective linear isometries of X. The group G is compact in $\mathcal{L}(X)$.

Assume, toward a contradiction, that X fails Property (P). Then for each n there exists a norm-one non-isometry $A_n \in \mathcal{L}(X)$ such that M_{A_n} is $1/n$-dense in S_X.

First, $\mathrm{dist}(A_n, G) \to 0$. Indeed, for every fixed $x \in S_X$, choose $y_n \in M_{A_n}$ with $\|x - y_n\| \le 1/n$. Since $\|A_n\| = 1$,

$$1 \ge \|A_n x\| \ge \|A_n y_n\| - \|A_n\| \, \|x - y_n\| \ge 1 - \frac{1}{n}.$$

Thus any operator-norm cluster point A of the sequence A_n satisfies $\|Ax\| = 1$ for every $x \in S_X$. Hence A is a surjective linear isometry, so $A \in G$. This proves $\mathrm{dist}(A_n, G) \to 0$.

For each n, choose $U_n \in G$ with $\|A_n - U_n\| = \mathrm{dist}(A_n, G)$, and put

$$B_n = U_n^{-1} A_n.$$

Then B_n is still a norm-one non-isometry, $M_{B_n} = M_{A_n}$, and

$$\mathrm{dist}(B_n, G) = \|B_n - I\|.$$

The equality follows because left multiplication by U_n^{-1} is an operator-norm isometry and maps G onto G.

Set

$$t_n = \|B_n - I\| > 0, \qquad C_n = \frac{B_n - I}{t_n}.$$

Passing to a subsequence, $C_n \to C$ in $\mathcal{L}(X)$, with $\|C\| = 1$.

We claim that for every $x \in S_X$,

$$\max_{f \in J(x)} f(Cx) = 0. \tag{1}$$

Fix $x \in S_X$. Let $f \in J(x)$. Since $\|B_n\| \le 1$,

$$f(B_n x) \le \|B_n x\| \le 1 = f(x).$$

As $B_n x = x + t_n C_n x$, this gives $f(C_n x) \le 0$. Letting $n \to \infty$,

$$f(Cx) \le 0 \qquad (f \in J(x)). \tag{2}$$

For the opposite inequality, choose $y_n \in M_{B_n}$ with $\|y_n - x\| \le 1/n$. Pick $g_n \in J(B_n y_n)$. Since $\|B_n y_n\| = 1$,

$$1 = g_n(B_n y_n) = g_n(y_n) + t_n g_n(C_n y_n).$$

Also $g_n(y_n) \le \|y_n\| = 1$, so $g_n(C_n y_n) \ge 0$. After passing to a subsequence, $g_n \to g \in S_{X^*}$. Because $B_n y_n \to x$ and $g_n(B_n y_n) = 1$, we have $g(x) = 1$, hence $g \in J(x)$. Taking limits gives $g(Cx) \ge 0$. Together with (2), this proves (1).

Now fix $z \ne 0$ and define

$$h_z(s) = \left\| e^{sC} z \right\| \qquad (s \ge 0).$$

The vector $e^{sC} z$ is never zero. By the directional derivative lemma,

$$(h_z)'_+(s) = \max_{f \in J(e^{sC} z)} f(C e^{sC} z).$$

Writing $e^{sC} z = \alpha x$ with $\alpha > 0$ and $x \in S_X$, the right side equals

$$\alpha \max_{f \in J(x)} f(Cx) = 0$$

by (1). The zero right-derivative lemma therefore gives

$$\left\| e^{sC} z \right\| = \|z\| \qquad (z \in X,\ s \ge 0).$$

So $e^{sC} \in G$ for every $s \ge 0$.

Finally, since $e^{t_n C} \in G$,

$$t_n = \operatorname{dist}(B_n, G) \le \left\| B_n - e^{t_n C} \right\|.$$

But $B_n = I + t_n C_n$, $C_n \to C$, and

$$e^{t_n C} = I + t_n C + O(t_n^2),$$

so

$$\left\| B_n - e^{t_n C} \right\| \le t_n \|C_n - C\| + O(t_n^2) = o(t_n).$$

This contradicts $t_n > 0$. Therefore X satisfies Property (P). □

Scope and limitations

This proves the finite-dimensional half of the source question. The infinite-dimensional case remains open here: the proof uses compactness of the operator unit ball to obtain $A_n \to G$, compactness of G to choose nearest isometries, and compactness of S_{X^*} to pass to limiting supporting functionals.

The source paper assumes real scalars throughout. A complex version should follow after passing to the underlying real Banach space and interpreting supporting functionals as real supporting functionals, but that is not needed for the stated source question and is not claimed here.

Verification notes

- The source paper already proves the polyhedral case and special ℓ_p^2 cases. This proof handles arbitrary finite-dimensional real norms, including nonsmooth nonpolyhedral norms.
- The previous smooth finite-dimensional packet is preserved in the history folder; the present proof removes the smoothness hypothesis by using all supporting functionals.

- The only convex-analysis input is the standard formula for the one-sided directional derivative of a finite-dimensional norm.
- The nearest-isometry normalization is essential. It turns the limiting first-order object into an infinitesimal isometry and then contradicts the minimality identity $\mathrm{dist}(B_n, G) = \|B_n - I\|$.
- Bounded novelty checks were run against the local run indexes for arXiv:2407.07490, Property (P), uniform BPB/Bishop-Phelps-Bollobas keywords, and close phrases. A web search on 2026-06-19 for exact and near-exact phrases did not reveal a later answer.

Human review recommendation

Review as a likely valid full solution to the finite-dimensional part of the source question. The main audit points are the subdifferential claim (1), the right-derivative argument for e^{sC}, and the nearest-isometry normalization. These are all local, finite-dimensional steps.

Reference

D. Sain, A. Mal, K. Mandal and K. Paul, *On uniform Bishop-Phelps-Bollobas type approximations of linear operators and preservation of geometric properties*, arXiv:2407.07490.